

DEEP LEARNING — 046211
FINAL EXAM A
30/4/2022

EXAM DURATION: 3 HOURS

INSTRUCTIONS.

1. You can use a calculator and the 2 formula sheets (4 pages).
2. You need to answer all question parts, unless you received a special permit.
3. The weight of each question is written in the form. The total number of points is 100 points.
4. If you received a special permit (due to reserve duty) you can choose to answer only two questions out of three. In that case, the questions points will be normalized so the total number of points will remain 100.
5. Please write clearly and orderly. You can write either in English or in Hebrew.
6. You must give the full details of the solution derivation. Solutions without explanations will not be given a score, unless written otherwise.
7. Once you finish the exam, you need to use gradescope to upload your exam. In addition, you need to submit the notebook.

1. TRANSFORMERS

For a series of T values, each with d features $X \in \mathbb{R}^{T \times d}$, a Self-Attention (SA) layer with parameters

$$\mathbf{W}_Q \in \mathbb{R}^{d \times D}, \mathbf{W}_K \in \mathbb{R}^{d \times D}, \mathbf{W}_V \in \mathbb{R}^{d \times M}$$

is defined as:

$$\begin{aligned} (1) \quad & \mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \mathbf{K} = \mathbf{X}\mathbf{W}_K, \mathbf{V} = \mathbf{X}\mathbf{W}_V \\ (2) \quad & \text{SA}(\mathbf{X}) = \mathbf{V}' = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}} \right) \mathbf{V} \end{aligned}$$

We will now look at an Encoder-Decoder Transformer

1. Give an example usage for this kind of architecture.
2. Draw the architecture and shortly explain each of the following components: Positional Encoding, AddNorm, Feed Forward.
3. Explain the difference between the Encoder and Decoder components.

One of the issues in computing SA is the computation speed and the memory required for this computation.

4. How many multiplications are required for calculation of a single SA layer, as defined in Eq. 2 (without the computation needed for Eq. 1, and other non-multiplication operations).

5. We will now see how to make this computation more efficient. We define a similarity function between the vectors q and k as follows:

$$\text{sim}(\mathbf{q}, \mathbf{k}) = \exp \left(\frac{\mathbf{q}^T \mathbf{k}}{\sqrt{D}} \right).$$

Show that the expression for the i^{th} value in the series ($1 \leq i \leq T$):

$$(3) V'_i = \frac{\sum_{j=1}^T \text{sim}(\mathbf{Q}_i, \mathbf{K}_j) V_j}{\sum_{r=1}^T \text{sim}(\mathbf{Q}_i, \mathbf{K}_r)}$$

is equivalent to Eq. 2 above, where a matrix with index i denotes the i^{th} row in the matrix. For example, Q_i denotes the vector of the i^{th} row of the matrix Q .

6. Assuming we can write the similarity function as

$$\text{sim}(\mathbf{q}, \mathbf{k}) = \phi(\mathbf{q})^\top \phi(\mathbf{k})$$

show how we can re-write (3) while using the associativity rule for matrices

$$\left(\phi(\mathbf{Q}) \phi(\mathbf{K})^\top \right) \mathbf{V} = \phi(\mathbf{Q}) \left(\phi(\mathbf{K})^\top \mathbf{V} \right)$$

so we can reduce the amount of multiplications in calculating expression (3), by computing certain expression only once in the beginning. How many multiplication are required for compute each row in a SA layer this method? You can neglect any multiplication in the denominator of Eq. (3) in this part and the next (why?).

7. Under which conditions the number of multiplications in part 6 is smaller than in part 4?

2. OPTIMIZATION

We wish to optimize a loss function $\mathcal{L}(\mathbf{w})$, a twice differentiable function where $\mathbf{w}$ is a weight vector in $\mathbb{R}^d$ using Gradient Descent (GD) with some step size schedule $\eta_t > 0$

$$\forall t = 0, 1, \dots : \mathbf{w}(t+1) = \mathbf{w}(t) - \eta_t \nabla \mathcal{L}(\mathbf{w}(t))$$

assume that $\mathcal{L}(\mathbf{w}) \geq 0$ and define

$$\beta(\mathbf{w}, \eta) \triangleq \max_{\alpha \in [0,1]} \lambda_{\max}(\nabla^2 \mathcal{L}(\mathbf{w} - \alpha \eta \nabla \mathcal{L}(\mathbf{w}))) ,$$

where $\lambda_{\max}$ denotes the maximal eigenvalue.

- (1) Prove that if $\eta_t < \frac{2}{\beta(\mathbf{w}(t), \eta_t)}$, then the loss decreases monotonically at step t .

Hint: From Taylor Expansion

$$\mathcal{L}(\mathbf{w}(t) - \eta_t \nabla \mathcal{L}(\mathbf{w}(t))) \leq \mathcal{L}(\mathbf{w}(t)) - \eta_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 + \frac{1}{2} \eta_t^2 \beta(\mathbf{w}(t), \eta_t) \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2$$

From here forward, for simplicity we approximate $\beta(\mathbf{w}, \eta_t) \approx \beta(\mathbf{w}) = \lambda_{\max}(\nabla^2 \mathcal{L}(\mathbf{w}))$.

- (2) Find a step size schedule η_t (as a function of $\beta(\mathbf{w}(t))$) for which we are guaranteed the fastest loss decrease at step t . Hint: Use the analysis from the previous part.
- (3) Prove that if $\eta_t = \frac{1}{\beta(\mathbf{w}(t))} \left(1 - \sqrt{1 - \frac{1}{a_t}}\right)$ for some diverging series (for which $\sum_{t=0}^{\infty} a_t = \infty$) then we converge to a stationary point.
- (4) Suppose $\beta = \max_{\mathbf{w} \in \mathbb{R}^d} \beta(\mathbf{w}) < \infty$. Do we have to decrease the step size for GD to converge to a stationary point? What about SGD? For what step schedule η_t are we guaranteed to converge for SGD (write an explicit mathematical condition)? No need to prove.
- (5) Write two step-size schedules are typically used in practice when training using SGD.

3. “CLASSICAL” GENERALIZATION IN NEURAL NETWORKS

We examine binary classification problem on $\mathcal{S} = \{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N$, a dataset of samples, with each $\mathbf{x} \in \mathbb{R}^d$ and $y \in \{-1, 1\}$ sampled i.i.d. $(\mathbf{x}, y) \sim P_{X,Y}$, and a function $f \in \mathcal{F}$ that perfectly fits the data ($\forall n : f(\mathbf{x}^{(n)}) = y^{(n)}$), where $\mathcal{F}$ is some pre-determined finite set functions (of size $|\mathcal{F}|$). In class we learned the following “classical” generalization bound:

Theorem. For any $f \in \mathcal{F}$, with probability $1 - \delta$,

$$\epsilon \triangleq \mathbb{P}_{(\mathbf{x}, y)}(f(\mathbf{x}) \neq y) < \frac{\log |\mathcal{F}| + \log 1/\delta}{N}$$

- (1) Explain in words: what does ϵ mean?
- (2) The Theorem says “with probability $1 - \delta$ ”. What is the random variable (or process) over which we have this probability?
- (3) Suppose $\mathcal{F}$ is the class of neural networks, with $k > N$ parameters, where each parameter is quantized to one of $q > 1$ values. Show this Theorem is not useful in this case.

Next, we aim prove the Theorem. We do this in three steps. Recall we defined

$$B_\epsilon(\mathcal{F}) = \{f \in \mathcal{F} : P_{X,Y}(f(\mathbf{x}) \neq y) > \epsilon\}$$

- (4) Prove that $\forall f \in B_\epsilon(\mathcal{F})$:

$$P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N} \stackrel{i.i.d.}{\sim} P_{X,Y} (\forall n : f(\mathbf{x}^{(n)}) = y^{(n)}) < (1 - \epsilon)^N.$$

- (5) Prove that (Hint: Union bound)

$$P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N} \stackrel{i.i.d.}{\sim} P_{X,Y} (\exists f \in B_\epsilon(\mathcal{F}) : \forall n : f(\mathbf{x}^{(n)}) = y^{(n)}) < e^{-\epsilon N} |\mathcal{F}|.$$

- (6) Combine sections 4-5 together to obtain the Theorem.

Exam Solution

QUESTION 1 SOLUTION

See solution to Question 1 in Spring 2023 Exam B.

QUESTION 2 SOLUTION

1. The proof is almost identical to the one from the lecture, except β now depends on t . Specifically, doing a Taylor expansion

$$\begin{aligned}\mathcal{L}(\mathbf{w}(t+1)) &= \mathcal{L}(\mathbf{w}(t) - \eta_t \nabla \mathcal{L}(\mathbf{w}(t))) \\ &\leq \mathcal{L}(\mathbf{w}(t)) - \eta_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 + \frac{1}{2} \eta_t^2 \beta(\mathbf{w}(t), \eta_t) \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 \\ &= \mathcal{L}(\mathbf{w}(t)) - \eta_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 \left(1 - \frac{1}{2} \eta_t \beta(\mathbf{w}(t), \eta_t)\right)\end{aligned}$$

since $0 < \eta_t < \frac{2}{\beta(t)}$, we have $c_t \triangleq 1 - \frac{1}{2} \eta_t \beta(\mathbf{w}(t), \eta_t) > 0$, so

$$(3.1) \quad \mathcal{L}(\mathbf{w}(t+1)) = \mathcal{L}(\mathbf{w}(t)) - \eta_t c_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 < \mathcal{L}(\mathbf{w}(t)),$$

i.e. the loss decrease monotonically.

2. The loss decreases the most in eq. 3.1 if $\eta_t c_t$ is the largest possible, i.e. we need to maximize

$$\eta^*(\mathbf{w}) = \operatorname{argmax}_{\eta} \eta - \frac{1}{2} \eta^2 \beta(\mathbf{w}, \eta)$$

taking the derivative we get the equation

$$1 - \eta \beta(\mathbf{w}, \eta) - \frac{1}{2} \eta^2 \frac{\partial \beta(\mathbf{w}, \eta)}{\partial \eta} = 0.$$

Approximating $\beta(\mathbf{w}, \eta) \approx \beta(\mathbf{w})$ we get

$$1 - \eta \beta(\mathbf{w}) = 0 \Rightarrow \eta^*(\mathbf{w}) = \frac{1}{\beta(\mathbf{w})}.$$

3. Summing eq. 3.1 we get a the telescopic sum on the one hand

$$\sum_{t=0}^T \mathcal{L}(\mathbf{w}(t+1)) - \mathcal{L}(\mathbf{w}(t)) = \mathcal{L}(\mathbf{w}(T+1)) - \mathcal{L}(\mathbf{w}(0)) \geq -\mathcal{L}(\mathbf{w}(0)) > -\infty,$$

where we used the fact $\mathcal{L}(\mathbf{w}(T+1)) \geq 0$ and on the other hand

$$\sum_{t=0}^T \mathcal{L}(\mathbf{w}(t+1)) - \mathcal{L}(\mathbf{w}(t)) = - \sum_{t=0}^T \eta_t c_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2.$$

Therefore

$$(3.2) \quad \sum_{t=0}^{\infty} \eta_t c_t \|\nabla \mathcal{L}(\mathbf{w}(t))\|^2 < \infty$$

to get to a stationary point, we need that $\|\nabla \mathcal{L}(\mathbf{w}(t))\| \rightarrow 0$, to guarantee that, we can require that

$$\sum_{t=0}^{\infty} \eta_t c_t < \infty$$

then we must have $\|\nabla \mathcal{L}(\mathbf{w}(t))\| \rightarrow 0$ for the sum to be bounded. So, we require that for some diverging series a_t

$$\sum_{t=0}^{\infty} a_t = \infty$$

we have that

$$a_t = \eta_t c_t = \eta_t - \frac{1}{2} \eta_t^2 \beta(\mathbf{w})$$

So

$$\eta_t^2 - \frac{2}{\beta(\mathbf{w})} \eta_t + \frac{1}{a_t \beta(\mathbf{w})} = 0$$

which implies

$$\eta_t = \frac{1}{\beta(\mathbf{w})} \pm \sqrt{\frac{1}{\beta^2(\mathbf{w})} - \frac{1}{a_t \beta(\mathbf{w})}} = \frac{1}{\beta(\mathbf{w})} \left(1 \pm \sqrt{1 - \frac{1}{a_t}} \right)$$

since we also must have $\eta_t < \frac{2}{\beta(\mathbf{w})}$ for $c_t > 0$, then we must take the solution with the minus sign.

4. For GD, for any constant

$$\eta < \frac{2}{\beta}$$

we will converge to a stationary point (as we learned in class, and as implied from previous sections here). For SGD, in some cases we can converge with a fixed step size, and in other cases we need to decay the step size. As we learned in class, to converge to a stationary point in SGD in general, we require

$$\sum_{t=1}^{\infty} \eta_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty.$$

5. Step decay, in which for some $\rho < 1$ and $T > 0$

$$\eta_t = \begin{cases} \rho \eta_t & , \text{if } t \bmod T = 0 \\ \eta_t & , \text{else} \end{cases}$$

and Cosine decay, in which for some $T > 0$

$$\eta_t = \frac{1}{2} \left(1 + \cos \left(\frac{t\pi}{T} \right) \right) \eta.$$

One can also increase the step size linearly during the beginning of training (step size warm up).

QUESTION 3 SOLUTION

1. It is the probability to have a classification error on a new unseen sample, i.e. the risk (which is typically estimated using the test error).
2. Probability over the random sampling of the training set $\mathcal{S}$ i.i.d from $P_{X,Y}$.
3. In this case, the number of possible parameter values is $|\mathcal{F}| = k^q$. Therefore the bound is

$$\epsilon < \frac{k \log q + \log(1/\delta)}{N}$$

since $k > N$ this upper bound is larger than 1, which means it is useless, as we already know that probability of classification error must be smaller or equal to 1 .

4. By definition $\forall f \in \mathcal{B}_\epsilon(\mathcal{F})$

$$P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N \sim P_{X,Y}} (f(\mathbf{x}^{(n)}) = y^{(n)}) < 1 - \epsilon.$$

Therefore, since we sample the datapoints i.i.d., we can take a product over all N points in $\mathcal{S}$:

$$P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N \sim P_{X,Y}} (\forall n : f(\mathbf{x}^{(n)}) = y^{(n)}) < (1 - \epsilon)^N.$$

5. We have

$$\begin{aligned} & P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N \sim P_{X,Y}} (\exists f \in \mathcal{B}_\epsilon(\mathcal{F}) : \forall n : f(\mathbf{x}^{(n)}) = y^{(n)}) \\ & < |\mathcal{B}_\epsilon(\mathcal{F})| (1 - \epsilon)^N \leq |\mathcal{F}| (1 - \epsilon)^N \leq |\mathcal{F}| e^{-\epsilon N} \end{aligned}$$

where in the first inequality we used the union bound, in the second $\mathcal{B}_\epsilon(\mathcal{F}) \subset \mathcal{F}$, and in the third $(1 - x) \leq e^{-x}$.

6. Denoting $\delta = P_{\{\mathbf{x}^{(n)}, y^{(n)}\}_{n=1}^N \sim P_{X,Y}} (\exists f \in \mathcal{B}_\epsilon(\mathcal{F}) : \forall n : f(\mathbf{x}^{(n)}) = y^{(n)})$ we get

$$\delta < |\mathcal{F}| e^{-\epsilon N}.$$

Taking the log of both sides

$$\log \delta < \log |\mathcal{F}| - \epsilon N$$

so

$$\epsilon < \frac{\log |\mathcal{F}| - \log \delta}{N} = \frac{\log |\mathcal{F}| + \log(1/\delta)}{N}.$$